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DEPARTMENT OF COMPUTER SCIENCE

Total Marks: 7.5

Obtained Marks:_____

Software Requirement Engineering-Lab

Project title: Online Food Delivery System

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1. Introduction

With the rapid development of technology and internet services, online food delivery platforms have become an essential part of modern urban life. These systems allow customers to browse restaurant menus, place orders online, and receive food deliveries without physically visiting restaurants. Online food delivery platforms provide convenience, save time, and allow customers to access a variety of restaurants from a single platform.

Despite the increasing popularity of these systems, many platforms still experience problems such as order delays, incorrect order processing, lack of proper order tracking, and communication issues between customers, restaurants, and delivery personnel. These problems often occur when software systems are developed without clearly defined and properly documented requirements.

Software Requirement Engineering is a critical process that helps identify, analyze, document, and manage system requirements before software development begins. It ensures that the needs of users and stakeholders are properly understood and translated into structured system requirements.

This project proposes to perform requirement engineering activities for an Online Food Delivery System. The focus of this project will not be the development of the full software application. Instead, the project will concentrate on identifying and organizing the system requirements using structured documentation and requirement management techniques.

The project will also demonstrate how requirement information can be tracked and organized using tools such as GitHub Issues and requirement documentation tables. Through this process, the project will illustrate how requirement engineering supports the development of reliable and efficient software systems.

2. Purpose of the Project

The purpose of this project is to apply Software Requirement Engineering principles in order to analyze and document the requirements of an Online Food Delivery System.

Many software failures occur because system requirements are unclear or incomplete during the early stages of development. Requirement engineering helps prevent such problems by ensuring that stakeholder needs are properly captured and documented.

This project will focus on requirement identification, requirement documentation, and

requirement organization. The project will demonstrate how requirements can be tracked and managed using simple tools such as GitHub issue tracking and structured requirement tables.

The final outcome of this project will be a detailed requirement proposal that describes the features and system behaviors expected from the proposed Online Food Delivery System.

3. Problem Statement

Online food delivery platforms are widely used but often encounter operational issues:

- Customers experience late or incorrect deliveries.
- Restaurants struggle to manage large volumes of online orders, especially during peak hours.
- Delivery personnel face difficulties communicating with customers and locating addresses.
- Existing systems often lack real-time order tracking.

These problems usually occur due to unclear or incomplete requirements. Without proper requirement engineering, developers may build systems that fail to meet user needs or quality standards.

This project aims to address these issues by performing a structured requirement engineering process, ensuring clarity, completeness, and proper management of system requirements.

4. Project Objectives

The main objective is to apply requirement engineering techniques to analyze and document system requirements for an Online Food Delivery System.

Specific Objectives

- Identify stakeholders and their needs
- Gather requirements through questionnaires
- Document and organize functional requirements
- Identify and categorize non-functional requirements
- Create a context diagram representing system interactions
- Demonstrate requirement management
- Provide a structured requirement document for future development

5. Project Scope

The scope of this project is limited to the requirement engineering phase of an Online Food Delivery System.

The project will include activities such as requirement identification, requirement documentation, and requirement organization. The project will focus on analyzing how the system should behave and what features it should include.

The system will be described in terms of the services it is expected to provide to customers, restaurants, and delivery personnel. However, the project will not involve the implementation or development of the complete software system.

Instead, the final output of the project will be a structured requirement proposal that can serve as a foundation for future system development

6. Stakeholder Identification

Stakeholders are individuals or groups that interact with or are affected by the system.

1) **Primary Stakeholders:**

- Customers – place orders and track delivery
- Restaurants – manage and confirm orders
- Delivery Personnel – receive and deliver orders
- System Administrators – manage system data and monitor performance
- Payment Providers / Banks – process online payments

2) **Secondary Stakeholders:**

- Notification Services – send updates to users
- Customer Support – handle queries and complaints
- Third-party APIs – integrate features like maps, ratings

Understanding stakeholder roles ensures that requirements capture all necessary system functionality and quality attributes.

7. Requirement Elicitation Plan

Requirement elicitation is the process of gathering information from stakeholders in order to understand their needs and expectations.

In this project, requirement elicitation will be performed through informal discussions and simple questionnaires with potential users who have experience using food

delivery applications.

Participants may be asked questions about their experiences with existing food delivery platforms. Questions may focus on common problems faced by users, features they consider important, and improvements they would like to see in such systems.

The responses collected from participants will help identify common requirements and expectations for the proposed system.

Requirement elicitation gathers information about what users expect from the system. In this project, elicitation is done through questionnaires and surveys with potential users.

Sample Questions:

1. What problems do you face while ordering food online?
2. Do you prefer real-time order tracking?
3. How important is delivery time?
4. Have you experienced incorrect orders?
5. What payment methods do you prefer?
6. Do you want communication with delivery personnel?
7. How important are notifications?
8. Do you prefer discounts and offers?
9. Is app performance important?
10. Would you like order history tracking?

Findings:

- Users prefer real-time tracking and accurate delivery
- Most reported delayed or incorrect orders
- Secure and multiple payment options are important
- Communication with delivery personnel is desired
- Users value fast and responsive interfaces

These insights form the basis for the functional and non-functional requirements.

8. Functional Requirements

Functional requirements describe what the system should do. The Online Food Delivery System must provide the following functionality:

ID	Functional Requirement
FR1	Users can register and create an account
FR2	Users can log in and log out
FR3	Users can browse restaurants
FR4	Users can view restaurant menus
FR5	Users can add items to the cart
FR6	Users can place an order
FR7	System sends order to restaurant
FR8	Restaurants can accept or reject orders
FR9	Users can track orders in real-time
FR10	Delivery personnel receive order assignments
FR11	Delivery personnel update delivery status
FR12	Users receive notifications for order updates
FR13	Users can make payments online
FR14	Admin manages users and system data
FR15	Users can view order history
FR16	Users can cancel orders
FR17	Users can rate and review services

9. Non-Functional Requirements

Non-functional requirements define system qualities such as performance, usability, and security.

Performance

- NFR1: System response time ≤ 2 seconds
- NFR2: Supports at least 1000 concurrent users

Security

- NFR3: Secure login and authentication
- NFR4: Encrypted payment data

Usability

- NFR5: Intuitive, user-friendly interface
- NFR6: Easy navigation

Reliability

- NFR7: 99% uptime

Scalability

- NFR8: Supports future growth

Availability

- NFR9: Available 24/7

Maintainability

- NFR10: Easy to update and maintain

10.Requirement Management

Requirement management ensures proper organization and tracking.

- Each requirement is assigned a unique ID
- Requirements are categorized as functional or non-functional
- Priority levels are set as high, medium, or low
- Structured requirement tables are maintained for reference
- Ensures that all requirements are clear, traceable, and updatable

11.Expected Outcome

The expected outcome is a complete requirement proposal for an Online Food Delivery System.

- Clear and organized functional and non-functional requirements
- A context diagram representing system interactions
- A foundation for system design and future development
- Demonstration of requirement management principles in practice

This project illustrates the value of requirement engineering in developing reliable, user-friendly, and efficient software systems.

12.Conclusion

Software Requirement Engineering is critical for developing systems that meet stakeholder expectations.

This project applies requirement engineering techniques to an Online Food Delivery System, covering elicitation, documentation, and management.

By preparing a structured requirement proposal, the project provides a solid foundation for future development and ensures that the system can be built efficiently and effectively.

13.Approval

We request approval for the Online Food Delivery System project as it provides a meaningful and practical way to apply software requirement engineering concepts learned in the course. The project demonstrates how structured requirement analysis, elicitation, documentation, and management can support the development of reliable and user-centric software systems.